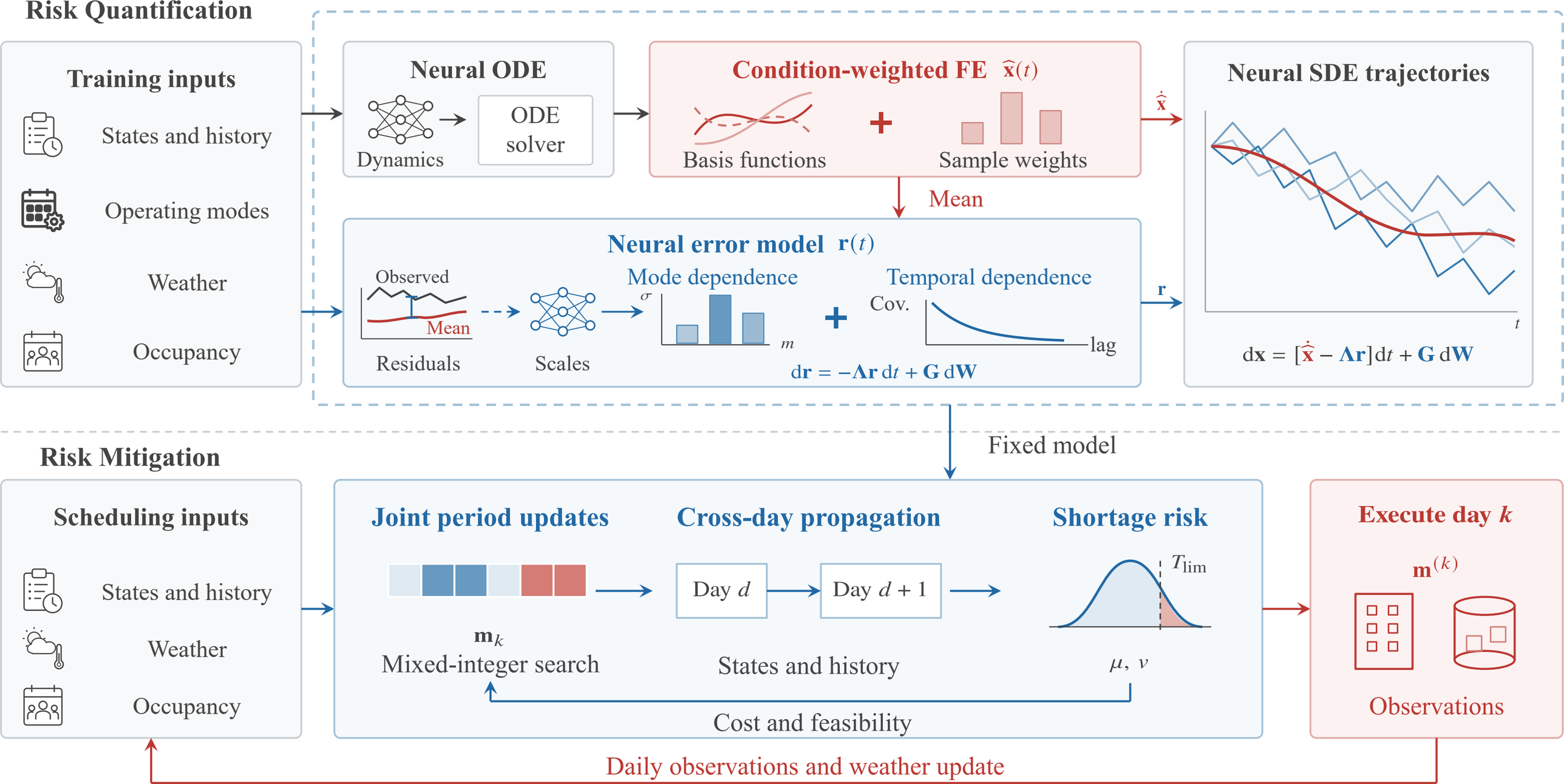

Risk Quantification
Training inputs
States and history
Operating modes
Weather
Occupancy
Neural ODE
Dynamics
ODE solver
Condition-weighted FE $\hat{\mathbf{x}}(t)$
Basis functions
Sample weights
Mean
Neural error model $\mathbf{r}(t)$
Observed
Mean
Residuals
Scales
Mode dependence
σ
m
Temporal dependence
Cov.
lag
$d\mathbf{r} = -\mathbf{\Lambda r}\,dt + \mathbf{G}\,d\mathbf{W}$
$\hat{\mathbf{x}}$
$\mathbf{r}$
Neural SDE trajectories
t
$d\mathbf{x} = [\dot{\hat{\mathbf{x}}} - \mathbf{\Lambda r}]dt + \mathbf{G}\,d\mathbf{W}$
Fixed model
Risk Mitigation
Scheduling inputs
States and history
Weather
Occupancy
Joint period updates
$\mathbf{m}_k$
Mixed-integer search
Cross-day propagation
Day d
Day $d+1$
States and history
Shortage risk
T_{lim}
$\mu,\ v$
Cost and feasibility
Execute day k
$\mathbf{m}^{(k)}$
Observations
Daily observations and weather update